

The Scales of Harmonies

An overview of 33 scales in 7 families, by **Oliver Prehn** (NewJazz) — colour-coded study sheet

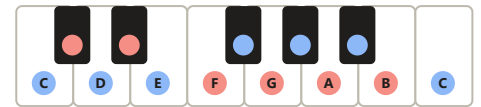
The 33 scales presented in the NewJazz lesson are organised in 7 families. Each family has one interval pattern; a scale is that pattern read from a given degree — which is why the rows of a family line up as a staircase. The 33 scales follow from a few simple rules:

- Only half, whole and whole-and-a-half steps
- Half steps may not be neighbours
- Whole and whole-and-a-half steps may not be neighbours
- Whole-and-a-half steps may not be neighbours

Following them limits the total to 33 scales, all well suited to building clear, evident harmonies.

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Reading the colours



• **Blue** and **red** are the two whole-tone sets of the keyboard. A **whole step (1)** stays in the same colour; a **half step (½)** or a **step and a half (1½)** crosses to the other. Each scale is a row of tiles, one per note, from the tonic to its octave: that two-colour sequence is the fingerprint of the mode. Colours are shown with the first scale of each family starting on C; down a family, each column is one and the same note. From a red tonic, swap the two colours. ■ A tile **split in two** is reached by a **step and a half**: the colour you come from on top, the one you arrive at below — not to be mistaken for a half step (families 3, 4 and 7).

FAMILY	POPULAR NAME	INTERVAL STEPS		*SYSTEMATIC NAME	CHORDS
1	Major	1-1-½-1-1-1-½	■ ■ ■ ■ ■ ■ ■	Ionian	Ma7
	Dorian	1-½-1-1-1-½-1	■ ■ ■ ■ ■ ■ ■	Dorian	mi7
	Phrygian	½-1-1-1-½-1-1	■ ■ ■ ■ ■ ■ ■	Phrygian	mi7
	Lydian	1-1-1-½-1-1-½	■ ■ ■ ■ ■ ■ ■	Lydian	Ma7
	Mixolydian	1-1-½-1-1-½-1	■ ■ ■ ■ ■ ■ ■	Mixolydian	7
	Natural minor	1-½-1-1-½-1-1	■ ■ ■ ■ ■ ■ ■	Aeolian	mi7
	Locrian	½-1-1-½-1-1-1	■ ■ ■ ■ ■ ■ ■	Locrian	mi7b5
2	Altered / Super Locrian	½-1-½-1-1-1-1	■ ■ ■ ■ ■ ■ ■	Ionian #1	7alt / mi7b5
	Ascending mel. minor	1-½-1-1-1-1-½	■ ■ ■ ■ ■ ■ ■	Dorian #7	miMa7
	Dorian b2	½-1-1-1-1-½-1	■ ■ ■ ■ ■ ■ ■	Phrygian #6	mi7
	Lydian Augmented	1-1-1-1-½-1-½	■ ■ ■ ■ ■ ■ ■	Lydian #5	Ma7#5
	Lydian dominant	1-1-1-½-1-½-1	■ ■ ■ ■ ■ ■ ■	Mixolydian #4	7
	Aeolian dominant	1-1-½-1-½-1-1	■ ■ ■ ■ ■ ■ ■	Aeolian #3	7
	Half diminished	1-½-1-1-½-1-1-1	■ ■ ■ ■ ■ ■ ■	Locrian #2	mi7b5
3	Major #5 / Major Aug.	1-1-½-1½-½-1-½	■ ■ ■ ■ ■ ■ ■	Ionian #5	Ma7#5
	Dorian #4	1-½-1½-½-1-½-1	■ ■ ■ ■ ■ ■ ■	Dorian #4	mi7
	Phrygian dominant	½-1½-½-1-½-1-1	■ ■ ■ ■ ■ ■ ■	Phrygian #3	7
	Lydian #2	1½-½-1-½-1-1-½	■ ■ ■ ■ ■ ■ ■	Lydian #2	Ma7
	Altered dominant bb7	½-1-½-1-1-½-1½	■ ■ ■ ■ ■ ■ ■	Mixolydian #1	dim7
	Harmonic minor	1-½-1-1-½-1½-½	■ ■ ■ ■ ■ ■ ■	Aeolian #7	miMa7
	Locrian #6	½-1-1-½-1½-½-1	■ ■ ■ ■ ■ ■ ■	Locrian #6	mi7b5
4	Harmonic Major	1-1-½-1-½-1½-½	■ ■ ■ ■ ■ ■ ■	Ionian b6	Ma7
	Dorian b5	1-½-1-½-1½-½-1	■ ■ ■ ■ ■ ■ ■	Dorian b5	mi7b5
	Phrygian b4	½-1-½-1½-½-1-1	■ ■ ■ ■ ■ ■ ■	Phrygian b4	mi7 / 7
	Lydian b3	1-½-1½-½-1-1-½	■ ■ ■ ■ ■ ■ ■	Lydian b3	miMa7
	Mixolydian b2	½-1½-½-1-1-½-1	■ ■ ■ ■ ■ ■ ■	Mixolydian b2	7
	Lydian augmented #2	1½-½-1-1-½-1-½	■ ■ ■ ■ ■ ■ ■	Aeolian b1	Ma7#5 / dim7
	Locrian bb7	½-1-1-½-1-½-1½	■ ■ ■ ■ ■ ■ ■	Locrian b7	dim7
5	Diminished	1-½-1-½-1-½-1-½	■ ■ ■ ■ ■ ■ ■	Diminished	dim7
	Dominant diminished	½-1-½-1-½-1-½-1	■ ■ ■ ■ ■ ■ ■	Inverted diminished	7
6	Whole tone	1-1-1-1-1-1	■ ■ ■ ■ ■ ■	Whole tone	7 #5/b5
7	Augmented	1½-½-1½-½-1½-½	■ ■ ■ ■ ■ ■	Augmented	Ma7
	Inverted Augmented	½-1½-½-1½-½-1½	■ ■ ■ ■ ■ ■	Inverted Augmented	6#5

*The systematic names do not follow the jazz-school habit of naming every scale after the Major scale. "Aeolian #7" is an Aeolian scale with a raised 7th step: Aeolian is the premise, #7 the alteration. The first 4 families thus run from Ionian to Locrian with a simple sequence of sharps or flats.

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